Department of Computer Science and Engineering
Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering
Course Code & Title: CB3491 Cryptography and Cyber Security
Name of the Faculty member: Mrs. P.Sabeena Burvin, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit V /** Authentication applications Kerberos
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Jig Saw
- **Date Implementation: 10.10.2025**
- **Justification:**

Kerberos is a cornerstone topic in Cryptography and Network Security as it demonstrates the practical application of symmetric key cryptography in real-world authentication systems. Understanding the Kerberos protocol equips students with the knowledge to Analyze multi-party authentication mechanisms used in enterprise environments. Understand the role of the Key Distribution Center (KDC) as a trusted authority. Apply ticket-based authentication concepts in modern security architectures. Evaluate the security strengths and weaknesses of symmetric-key based authentication and Appreciate protocol design challenges such as replay attacks, clock synchronization, and scalability.

Time Allotted for the Activity: 30 minutes

Details of the Implementation:

- After I completed the Kerberos authentication process, the students are given a group activity. To understand the role of the Key Distribution Center (KDC) as a trusted authority, the students are made to work into groups to discuss and security strengths and weaknesses of symmetric-key based authentication.
- The students are asked to identify different applications in various domains. Students were divided into 2 groups with 4 members in each group. Each student in a group is assigned with any one application. I delivered the instructions to the students on how to Apply ticket-based authentication concepts in modern security architectures, from each team one student is selected and framed as a team. Thus for each application there will be an expert group.
- The group members discuss within themselves and model the Kerberos authentication process for their application. After they have modelled the application, the expert group is collapsed and all the students are re-grouped into

their original group (initial team). Now each member in the team will be an expert in any one application. They shared their ideas to all the other members in the group thus enabling each student to learn all the applications. This enabled the students to get insight knowledge in this topic.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO9	PO12	PSO1
CO1	3	2	2	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO Mapping:

Innovative Practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO9**	**PO12**	**PSO1**
	3	2	2	2	1	1	1
Justification for correlation	Basic Computer Science Engineering and mathematics Knowledge for identifying various authentication process	Students will be able to analyze and define various types of autnentication	Students will design secure authentication solutions for network applications	Students will be able to use modern tools to perform identification process	Students will work individually and in teams to analyse Kerberos protocols	Written and oral communication with team members are improved	Students will apply Kerberos design skills in future security projects.

- **Images / Screenshot of the practice:**

Figure-1: Students working as a team in JIGSAW activity

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - Presenting Students felt the activity is very interesting. Students felt the activity conducted gave them a chance to improve their communication skills, also working as a team they can face group discussions in future work place. Through learning from peer members, students felt comfortable to clarify their doubts.
- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
  - Through this activity the students made to work as a team and explain to others also, like a group discussion activity. The students are made to understand the different the applications through knowledge sharing.
  - Students spent their time in self-learning.
- ***Challenges faced in implementation:***
  - The grouping of the students was difficult, since some students were not present. Initially 2 teams were planned with team size of 6 members, but the team size is reduced and it was difficult to form the expert group.
  - Some students in the team did not participate actively in the team.
  - Individual assessment/observation was difficult.

References:

- https://www.jigsaw.org/
- https://www.readingrockets.org/strategies/jigsaw
- https://www.youtube.com/watch?v=euhtXUgBEts
- https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/